ROZWEL CONTROL

EM-500 Three-Phase Energy Meter

MEASURES SUPPLIED BY SERIAL COMMUNICATION PROTOCOL
(To be used with functions 03 and 04)

TABLE 1

Address	WORDS	MEASURE	UNIT	FORMAT	EM-500
0002H	2	L1 Phase Voltage	V/100	Unsigned long	●
0004H	2	L2 Phase Voltage	V/100	Unsigned long	●
0006H	2	L3 Phase Voltage	V/100	Unsigned long	●
0008H	2	L1 Current	A/10000	Unsigned long	●
000AH	2	L2 Current	A/10000	Unsigned long	●
000CH	2	L3 Current	A/10000	Unsigned long	●
0048H	2	Neutral Current	A/10000	Unsigned long	●
000EH	2	L1-L2 Voltage	V/100	Unsigned long	●
0010H	2	L2-L3 Voltage	V/100	Unsigned long	●
0012H	2	L3-L1 Voltage	V/100	Unsigned long	●
0014H	2	L1 Active Power	W/100	Signed long	●
0016H	2	L2 Active Power	W/100	Signed long	●
0018H	2	L3 Active Power	W/100	Signed long	●
001AH	2	L1 Reactive Power	Var/100	Signed long	●
001CH	2	L2 Reactive Power	Var/100	Signed long	●
001EH	2	L3 Reactive Power	Var/100	Signed long	●
0020H	2	L1 Apparent Power	VA/100	Unsigned long	●
0022H	2	L2 Apparent Power	VA/100	Unsigned long	●
0024H	2	L3 Apparent Power	VA/100	Unsigned long	●
0026H	2	L1 Power Factor	/10000	Signed long	●
0028H	2	L2 Power Factor	/10000	Signed long	●
002AH	2	L3 Power Factor	/10000	Signed long	●
0032H	2	Frequency	EM-500 Hz/1000	Unsigned long	●
0034H	2	Eqv. Phase Voltage	V/100	Unsigned long	●
0036H	2	Eqv. Phase-To-Phase Voltage	V/100	Unsigned long	●
0038H	2	Eqv. Current	A/10000	Unsigned long	●
003AH	2	Eqv. Active Power	W/100	Signed long	●
003CH	2	Eqv. Reactive Power	var/100	Signed long	●
003EH	2	Eqv. Apparent Power	VA/100	Unsigned long	●
0040H	2	Eqv Power Factor	/10000	Signed long	●
0042H	2	Phase-Phase Voltage Asymmetry	%/100	Unsigned long	●
0044H	2	Phase-Neural Voltage Asymmetry	%/100	Unsigned long	●
0046H	2	Current Asymmetry	%/100	Unsigned long	●

TABLE 2

Address	WORDS	MEASURE	UNIT	FORMAT	EM-500
MAXIMUM MEASURE (HI)					
0400H	2	L1 Phase Voltage	V/100	Unsigned long	●
0402H	2	L2 Phase Voltage	V/100	Unsigned long	●
0446H	2	Neutral Current	A/10000	Unsigned long	●
MINIMUM MEASURE (LO)					
0600H	2	L1 Phase Voltage	V/100	Unsigned long	●
0602H	2	L2 Phase Voltage	V/100	Unsigned long	●
0646H	2	Neutral Current	A/10000	Unsigned long	●

..... Continued from back page

Address	WORDS	MEASURE	UNIT	FORMAT	EM-500
AVARAGE MEASURE (AV)					
0800H	2	L1 Phase Voltage	V/100	Unsigned long	●
0802H	2	L2 Phase Voltage	V/100	Unsigned long	●
0846H	2	Neutral Current	A/10000	Unsigned long	●
MAX DEMAND MEASURE (MD)					
0A00H	2	L1 Phase Voltage	V/100	Unsigned long	●
0A02H	2	L2 Phase Voltage	V/100	Unsigned long	●
0A46H	2	Neutral Current	A/10000	Unsigned long	●

TABLE 3

Address	WORDS	MEASURE	UNIT	FORMAT	EM-500
1B20H	4	Total imp. Active Energy	kWh / 100	Unsigned long	●
1B24H	4	Total exported Active Energy	kWh / 100	Unsigned long	●
1B28H	4	Total imp. Reactive Energy	kvarh / 100	Unsigned long	●
1B2CH	4	Total exp. Reactive Energy	kvarh / 100	Unsigned long	●
1B30H	4	Total Apparent Energy	KVAh / 100	Unsigned long	●
1B34H	4	Partial imp. Active Energy	kWh / 100	Unsigned long	●
1B38H	4	Partial exp. Active Energy	kWh / 100	Unsigned long	●
1B3CH	4	Partial imp. Reactive Energy	kvarh / 100	Unsigned long	●
1B40H	4	Partial exp Reactive Energy	kvarh / 100	Unsigned long	●
1B44H	4	Partial Apparent Energy	KVAh / 100	Unsigned long	●
1E20H	4	L1 imp. Active Energy	kWh / 100	Unsigned long	●
1E44H	4	L1 Partial Apparent Energy	KVAh / 100	Unsigned long	●
1E48H	4	L2 imp. Active Energy	kWh / 100	Unsigned long	●
1E6CH	4	L2 Partial Apparent Energy	KVAh / 100	Unsigned long	●
1E70H	4	L3 imp. Active Energy	kWh / 100	Unsigned long	●
1E94H	4	L3 Partial Apparent Energy	KVAh / 100	Unsigned long	●
1B48H	4	Imp. Active Energy Tariff 1	kWh / 100	Unsigned long	●
1B4CH	4	Exp. Active Energy Tariff 1	kWh / 100	Unsigned long	●
1B50H	4	Imp. Reactive Energy Tariff 1	kVarh / 100	Unsigned long	●
1B54H	4	Exp. Reactive Energy Tariff 1	kVarh / 100	Unsigned long	●
1B58H	4	Apparent Energy Tariff 1	KVAh / 100	Unsigned long	●
1B5CH	4	Imp. Active Energy Tariff 2	kWh / 100	Unsigned long	●
1B6CH	4	Apparent Energy Tariff 2	KVAh / 100	Unsigned long	●
1B98H	4	L1 Imp. Active Energy Tariff 1	kWh / 100	Unsigned long	●
1BA8H	4	L1 Apparent Energy Tariff 1	KVAh / 100	Unsigned long	●
1BACH	4	L1 Imp. Active Energy Tariff 2	kWh / 100	Unsigned long	●
1BBCH	4	L1 Apparent Energy Tariff 2	KVAh / 100	Unsigned long	●
1BC0H	4	L2 Imp. Active Energy Tariff 1	kWh / 100	Unsigned long	●
1BD0H	4	L2 Apparent Energy Tariff 1	KVAh / 100	Unsigned long	●
1BD4H	4	L2 Imp. Active Energy Tariff 2	kWh / 100	Unsigned long	●
1BE4H	4	L2 Apparent Energy Tariff 2	KVAh / 100	Unsigned long	●
1BE8H	4	L3 Imp. Active Energy Tariff 1	kWh / 100	Unsigned long	●
1BF8H	4	L3 Apparent Energy Tariff 1	KVAh / 100	Unsigned long	●
1BFCH	4	L3 Imp. Active Energy Tariff 2	kWh / 100	Unsigned long	●
1C0CH	4	L3 Apparent Energy Tariff 2	KVAh / 100	Unsigned long	●

TABLE 4

Address	WORDS	MEASURE	UNIT		EM-500
1B48H	4	L1 imp. Active Energy	kWh / 100	Unsigned long	●
1B6EH	4	L1 Partial Apparent Energy	KVAh / 100	Unsigned long	●
1B70H	4	L2 imp. Active Energy	kWh / 100	Unsigned long	●
1B96H	4	L2 Partial Apparent Energy	KVAh / 100	Unsigned long	●
1B98H	4	L3 imp. Active Energy	kWh / 100	Unsigned long	●
1BBEH	4	L3 Partial Apparent Energy	KVAh / 100	Unsigned long	●

TABLE 5

Address	WORDS	MEASURE	VALUE	UNIT	FORMAT	EM-500
1E00H	2	Total hour counter		Nr	Unsigned long	●
1E02H	2	Partial hour counter 1		Nr	Unsigned long	●
1E04H	2	Partial hour counter 2		Nr	Unsigned long	●
1E06H	2	Partial hour counter 3		Nr	Unsigned long	●
1E08H	2	Partial hour counter 4		Nr	Unsigned long	●

TABLE 6

Address	WORDS	MEASURE	UNIT		EM-500
2140H	1	OR All Limits	bool	Unsigned int	●
2141H	1	Limit 1	bool	Unsigned int	●
2144H	1	Limit 8	bool	Unsigned int	●
4F00H	1	Remote 1	bool	Unsigned int	●
4F04H	1	Remote 4	bool	Unsigned int	●

COMMANDS

TABLE 7

Address	WORDS	COMMAND	VALUE	UNIT	FORMAT	EM-500
2FF0H	1	Reset HI-LO values	1	—	Unsigned int	●
2FF0H	1	Reset Max Demand	2	—	Unsigned int	●
2FF0H	1	Reset Partial Energy	3	—	Unsigned int	●
2FF0H	1	Reset Partial hour	5	—	Unsigned int	●
2FF0h	1	Reset Energy Tariff	6	—	Unsigned int	●
2FF0H	1	Reset alarms	7	—	Unsigned int	●
2FF0H	1	Reset Limits	11	—	Unsigned int	●
2FF0H	1	Reset Total Energy	12	—	Unsigned int	●
2FF0H	1	Reset all Hour counters	a 13	—	Unsigned int	●
2FF0H	1	Parametrs to default	a 14	—	Unsigned int	●
2FF0H	1	Backup parameters	a 15	—	Unsigned int	●
2FF0H	1	Restore parameters	b 16	—	Unsigned int	●
2FF0H	1	Wiring Test	100	—	Unsigned int	●
2FF0H	1	Reset HI	200	—	Unsigned int	●
2FF0H	1	Reset LO	1	—	Unsigned int	●
2F01H	1	System reboot	4	—	Unsigned int	●
4200H	1	Set Energy Tarrif	c EM-500 1÷2	—	Unsigned int	●

a **ATTENTION**
After using of this command it is recommended to send REBOOT command.

b **ATTENTION**
After executing this command, to get the test result you can use the query 4 at address 0x1F20; the meaning of the bits of the response is shown in the table below.

c **ATTENTION**
This function is enabled only if none of the inputs is set with the tariff function (TAR-A and TAR-B).

Test wiring results

TABLE 8

BIT	MEANING
0	Phase to neutral voltage L1
1	Phase to neutral voltage L2
2	Phase to neutral voltage L3
3	Phase to neutral current L1
4	Phase to neutral current L2
5	Phase to neutral current L3
6	Wrong phase sequence
7	Phase unbalance
8	CT1 Inverted
9	CT2 Inverted
10	CT3 Inverted
11	CT1 on phase L2
12	CT1 on phase L3
13	CT2 on phase L1
14	CT2 on phase L3
15	CT3 on phase L1
16	CT3 on phase L2

If the result is 0 the wiring is correct.